Venkata Lakshmi Parimala Pasupuleti

AI Machine Learning Engineer · Agentic AI · Multi-Agent Orchestration · RAG · LangGraph · Workflow Automation · Python · LLMs · AWS · Azure

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Virginia, USA — Open to Remote / Hybrid

PROFESSIONAL SUMMARY

AI/ML Engineer with 9+ years of enterprise software engineering and 3+ years hands-on **Agentic AI** development — designing and deploying **multi-agent systems**, **RAG pipelines**, **LLM orchestration**, and **operations-focused workflow automation** in production. Built **AgentProbe**: an enterprise multi-agent LangGraph runtime (Orchestrator → RAG Tracer → Auto Fix) with inter-agent state handoffs, tool calling, and multi-provider LLM routing (Anthropic, OpenAI, Groq, Ollama) — production-ready agentic AI architecture with LLM-as-judge model evaluation. Built **Leasel**: an end-to-end **Intelligent Document Processing (IDP)** and agentic RAG pipeline achieving **93.6% precision** on 1,240 enterprise documents. Eliminated **80% of manual operational workload** through enterprise AI automation (Python, PyTorch, OpenAI API) across **1.6M financial records**. Expert Python and SQL; AWS + Azure cloud; ChromaDB, FAISS-compatible vector databases; Docker; CI/CD. M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026).

WORK EXPERIENCE

Magizhs Tech Solution
May 2025 – Aug 2025

AI/ML Engineer — Agentic AI · Workflow Automation · RAG · Operations AI

- Designed and built an **agentic AI workflow automation** system as a **LangGraph** stateful graph (parse → plan → retrieve → validate → conditional routing) — **multi-agent orchestration** with inter-agent handoffs, **tool calling**, and **prompt engineering** translating natural-language business requirements into structured operational plans; collaborated cross-functionally with operations SMEs to map and automate complex workflows.
- Implemented a **RAG** knowledge base (ChromaDB-compatible retrieval over regulatory corpus) grounding every AI decision in cited sources; designed **guardrails** and validation layers enforcing operational constraints before execution — operations-focused AI delivering production-ready, enterprise-grade compliance outcomes.

George Mason University
Aug 2024 – Aug 2025

Research Associate — AI/ML Modeling · Model Evaluation · Analytics

- Built multi-source prediction models (Python, R) achieving **$R^2 = 0.977$** across 190+ countries; applied model evaluation frameworks, hypothesis testing, and cross-validation; surfaced anomalies across 3 pipelines using **Isolation Forest**; designed strategic dashboards (Tableau, ggplot2) improving comprehension **40%**.

Entrata
Jan 2017 – Aug 2024

Senior Software Engineer — Enterprise AI Pipelines · Workflow Automation · AWS Cloud · Full-Stack

- Designed and deployed **enterprise AI pipeline** processing **1.6M financial records** (Python, PyTorch, OpenAI API) — automated operational workflows replacing manual business processes; implemented **business process analysis** and **workflow mapping** to translate operations requirements into scalable AI solutions; eliminated **80% of manual operational workload** with zero compliance violations.
- Built and maintained **REST APIs** and microservices (PHP, Python, PostgreSQL, RabbitMQ) for 100K+ monthly active users at 99.5% uptime; architected **AWS** cloud infrastructure (S3, EC2, RDS, Glue); led **microservices architecture** redesign (Docker, service-based — 50% scalability gain); optimized **200+ SQL queries** (30% performance gain); built **third-party integrations** (Salesforce, API connections, CRM pipelines).
- Built **CI/CD pipelines** (GitHub Actions); conducted 300+ peer code reviews; mentored 5+ engineers; delivered cross-functional collaboration across compliance, operations, product, and engineering stakeholders across 120+ Agile sprints.

PROJECTS

AgentProbe: Enterprise Multi-Agent AI Testing Toolkit — LangGraph · A2A · LLM-as-Judge

Python · LangGraph · LangChain · FastAPI · ChromaDB · Anthropic · OpenAI · Groq · Ollama · React · Docker · GitHub Actions — github.com/venkatalakshmiparimala/AgentProbe

- Built a 3-agent **LangGraph** runtime (Orchestrator → RAG Tracer → Auto Fix) with inter-agent state handoffs, **multi-agent orchestration**, tool calling, and **multi-provider LLM routing** (Anthropic, OpenAI, Groq, Ollama) — 15-endpoint **FastAPI REST API** backend; 5-metric **LLM-as-judge** model evaluation with **CI/CD** (GitHub Actions); **ChromaDB** vector knowledge base with agent observability, tracing, and retrieval quality dashboards.

Leasel: Intelligent Document Processing & Agentic RAG Pipeline — 93.6% Precision

Python · LangGraph · RAG · ChromaDB · Scikit-learn · SQLite · reportlab — github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built an enterprise **Intelligent Document Processing (IDP)** and agentic **RAG** pipeline across 1,240 documents (5 US states + federal standards) — **prompt engineering**, LLM-as-judge evaluation (pass@1 81.7%, pass@3 88.3%), deterministic rules engine, full data lineage; achieved **93.6% precision, 87.9% F1** across 10 compliance rules; production-ready, audit-ready, and fully traceable.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — Graduated May 2026

GPA: 3.87 · Fairfax, Virginia · Machine learning, agentic AI, RAG pipelines, model evaluation, cloud AI, workflow automation

B.Tech Computer Science & Engineering — Jawaharlal Nehru Technological University, Anantapur — 2013 – 2017

TECHNICAL SKILLS

Agentic AI & LLMs: Agentic AI, Multi-Agent Orchestration, LangChain, LangGraph, LlamaIndex (familiar), Semantic Kernel (familiar), OpenAI, Anthropic, Azure OpenAI, Groq, Ollama, RAG Pipelines, Prompt Engineering, Tool Calling, Function Calling, Guardrails, LLM-as-Judge Model Evaluation, Intelligent Document Processing (IDP), Workflow Automation, Operations AI

Cloud, Data & APIs: AWS (S3, EC2, RDS, Glue), Azure, REST APIs (FastAPI), Docker, CI/CD (GitHub Actions), Microservices, PostgreSQL, MySQL, ChromaDB (vector DB), FAISS-compatible, Embeddings, Semantic Search, Python (expert), SQL (expert — 200+ production optimizations), R, JavaScript, PHP, Bash, Linux, Git, Agile/Scrum (120+ sprints)

ML & Analytics: PyTorch, Scikit-learn, Isolation Forest, Random Forest, XGBoost, BERT, spaCy, Hugging Face, NLP, Feature Engineering, Statistical Modeling, Tableau, Power BI, ggplot2